

INTRODUCTION

Agriculture remains the cornerstone of India's economy, supporting nearly half of the population and providing employment, nutrition, and rural development. The Kharif season, spanning from June to October, is particularly vital due to its reliance on the south-west monsoon. Approximately 55% of India's net sown area is rainfed, making the timely and equitable distribution of monsoon rains crucial for agricultural success.

In 2025, the India Meteorological Department (IMD) forecasts an above-normal monsoon season, with cumulative rainfall expected at 105% of the Long Period Average (LPA) of 87 cm. The monsoon's onset is anticipated to be slightly delayed in southern India but should reach Maharashtra by May 27, facilitating timely sowing preparations. The IMD's forecast of an above-normal monsoon offers a promising outlook but still carries risks of intra-seasonal dry spells and localized flooding. The Indian government's initiatives, such as the "Panchayat Mausam Seva," aim to deliver hyper-local weather forecasts at the panchayat level, ensuring that farmers receive accurate and timely information.

The Kharif season witnesses the cultivation of a diverse array of crops across India's agro-climatic zones, including rice, maize, cotton, soybean, groundnut, bajra, jowar, tur, urad, and moong. Despite advancements in agricultural technology, the sector remains vulnerable to climatic fluctuations such as delayed rains, uneven distribution, and extreme weather events. These challenges can adversely affect crop yields, farm incomes, and rural livelihoods.

To mitigate these risks, it is imperative to provide farmers with location-specific, research-backed guidance. Agro-advisories play a pivotal role by offering timely, region-specific, science-based information on weather patterns, crop management, pest control, harvesting, and storage. These advisories integrate seasonal outlooks with crop calendars and state-wise sowing windows, enabling farmers to select appropriate crop varieties and sowing dates, optimize resource use, and prevent yield losses.

This advisory serves as a crucial tool for agricultural officers, Farmer Producer Organizations (FPOs), and NGOs involved in rural development. By bridging the information gap between research and practice, it empowers farmers with the knowledge needed to make informed decisions, thereby contributing to the vision of a self-reliant, prosperous, and inclusive India @ VIKSIT BHARAT 2047.

Soybean

Use improved high yielding soybean varieties such as JS 20-69, RVS 2002-4, JS 20-98, JS 20-116, Indore Soya 142, NRC 150, NRC 152, JS 2172, RVSM 2011-35 and Indore Soya 130. To eradicate the pest propagules, including the pupae of major pests and the spore structures of major pathogens that remain on farms, such as on threshing floors and in the soil, it is essential to minimize the initial pest/pathogen load. Sanitation of threshing floors, deep ploughing of fields, and cleaning of bunds are necessary. Farmers should also collect all crop debris and bury it in a compost pit. Seed materials should be carefully processed to remove physically damaged, discolored, or undersized seeds. Only clean, healthy, pest- and disease-free material should be stored as seed. Also treat the seeds with Carbendazim 25% + Mancozeb 50% WS @ 3 g/kg seed + biological fungicide *Trichoderma viride* @ 8-10 g/kg seed, Bradyrhizobium culture and Phosphate Solubilizing Bacteria @ 10 g/kg seed each to get higher yields.

For the control of stem fly in soybean, treat seeds with the Thiamethoxam 30 FS @ 10 ml/kg seed before sowing. Ensure that soybean is sown on raised bed and Broad Bed and Furrow methods, to conserve moisture. For early maturing and short-statured varieties, maintain 30 cm row spacing. For long-duration and tall varieties, maintain 45 cm spacing between rows and 10 cm between plants. Apply fertilizers based on soil test recommendations. Generally, N: P²O⁵: K₂O at 25:60:40 kg/ha along with 20 kg/ha bentonite sulphur is recommended.

COMMERCIAL CROPS

Sugarcane

Both the nymphs and adults of whitefly suck the sap from the leaves, leading to stunted growth and a decrease in sugar content. To control this insect, spray Fenitrothion 50 EC at 2000 ml/ha at 15-day intervals. For control of leaf hopper, spray Chlorpyrifos 20% EC @ 1500 ml/ha or Dichlorvos 76% EC @ 375 ml/ha or Monocrotophos 36% SL @ 1000 ml/ha.

VEGETABLE CROPS

The current weather is favorable for onion cultivation. Before sowing, treat the seeds with Captan 75 WS @ 2.5 g/kg seed. In transplanted onion, dip the onion roots in a solution of Carbendazim 12% + Mancozeb 63% @ 2.5 g/L of water prior to transplanting. Raise the onion nursery with high-yielding varieties such as Agrifound Dark Red, Pusa Ridhi, N-53, Bhima Super, Bhima Raj, and Bhima Dark Red @ 12-15 kg seeds/ha. Considering the current temperature, farmers can begin garlic sowing. Ensure that the soil has adequate moisture before sowing. Recommended improved varieties to be sown such as G-1, G-41, G-50, and G-282.

During field preparation, apply the recommended dose of farmyard manure along with phosphorus fertilizer. Top dressing of nitrogen @ 20-25 kg/ha for cucurbit crops and 30-40 kg/ha for tomato, brinjal, and chili to be done for getting higher yield. For better fruit setting in tomato, brinjal, and chili, spray NAA @ 125 ml/ha. To prevent fruit cracking in tomatoes, spray 0.1% of borax at fruit setting stage.

For management of early blight and fruit rot of tomato, spray Copper Oxychloride @ 3 g/L or Metalaxyl 4% + Mancozeb 64% WP @ 1.5 g/L. For control of fruit and shoot borer

Nutrient management includes application of potassium at 50 kg/ha along with recommended nitrogen and phosphorus doses, and in saline soils, gypsum should be applied at 5 t/ha. To enhance crop nutrition, foliar spray of 0.2% boric acid or nano boron should be applied at 30, 45, and 60 DAS. For plant protection, collar and stem rot can be managed with a spray of Mancozeb or neem extract around 75 DAS. White grub infestation can be controlled through seed treatment with Chlorpyrifos or Imidacloprid, and soil application of *Metarhizium anisopliae* or Carbofuran. Sucking pests should be managed through seed treatment with Imidacloprid or foliar spray with Thiamethoxam, while defoliators can be controlled using Quinalphos, Chlorpyrifos, or Flubendiamide. Harvesting should be done at 120-125 DAS using a blade harrow, followed by threshing to separate the pods.



Groundnut Demo Plot

Soybean

Soybean grows best in well-drained loamy soils and should be avoided in saline or alkaline conditions. Deep ploughing and the application of farmyard manure (FYM) are essential for soil preparation. Recommended varieties include JS 335, GJS-3, and GS-4. Sowing is done in June-July with a spacing of 45×5 cm and a depth of 3-5 cm, using a seed rate of 70-80 kg/ha. Seeds should be treated with Carbendazim or Thiram along with *Rhizobium* and phosphate-solubilizing bacteria (PSB) to enhance germination and nutrient uptake.

For weed management, Pendimethalin is applied pre-emergence and Imazethapyr post-emergence, along with hand weeding at 20 and 40 days after sowing (DAS). To manage sucking pests, seed treatment with Imidacloprid is effective, while collar rot can be controlled by treating seeds with Fluxapyroxad or a combination of Carboxin and Thiram. In case of defoliator infestation, spraying Indoxacarb, Lambda-cyhalothrin, or a Novaluron mixture is recommended. For rust and leaf spot diseases, fungicides like Hexaconazole or Pyraclostrobin should be applied. Additionally, 75% of the recommended dose of fertilizers (RDF) should be supplemented with 25 kg/ha of Mycorrhiza and *Rhizobium* seed treatment to enhance soil health and crop productivity.

Cotton (Bt & Desi)

Cotton cultivation thrives best in well-drained loamy to sandy loam soils. Prior to sowing, deep ploughing is essential along with the incorporation of 10-15 t/ha of farmyard manure (FYM) to enhance soil fertility. Sowing is typically carried out in June-July under rainfed conditions and during April-May for irrigated crops. Recommended spacing for normal planting is 90-120 × 30-45 cm, while under high-density planting (HDP) systems, 60 × 30 cm spacing is advised. Suitable varieties include Bt hybrids like G. Cot. Hy. 24 and 26, and desi types such as GADC-3 and GADC-4. The seed rate varies by type, with 1.5-2 kg/ha for Bt varieties and 10-12 kg/ha for desi types.

To ensure the well-being of livestock, provide adequate shade, cool water, and a mineral mix for proper nutrition. Regularly shower animals to help keep them cool. Vaccinate for Hemorrhagic Septicemia (HS) and Black Quarter (BQ), and deworm young animals to maintain their health. For tick control, use Deltamethrin. Additionally, use sanitizers in the sheds to reduce the presence of flies and mosquitoes, promoting a clean and hygienic environment for the animals.

CENTRAL GUJARAT

This region of Gujarat includes Ahmedabad, Anand, Kheda, Vadodara, Panchmahal, Mahisagar, and Dahod. They receive 600-900 mm of rainfall annually. The region has shallow to medium-depth black and red soils, which are typically low in fertility and organic carbon.

Maize

For sowing in Central Gujarat, use improved varieties like Narmda Moti, GM-2, GM-4, and GM-6, and hybrids such as GAYMH-1 and GAWMH-2. The ideal sowing time is from June 15th to July 15th. Prepare the land with 1-2 harrowings and 1 planking, applying manure before sowing. For spacing, use 60x20 cm for improved varieties and 75x20 cm for hybrids. The seed rate is 20-25 kg/ha for improved varieties and 25-30 kg/ha for hybrids. Fertilizer application should be 100:50:00 (N:P:K) kg/ha for improved varieties and 120:60:00 (N:P:K) kg/ha for hybrids. For plant protection against Fall Armyworm, use light traps, pheromone traps, and Neem seed powder/oil spray.

Rice

For transplanted rice cultivation, use 20-25 days old seedlings, with 2-3 plants per hill. Treat seedlings with *Azotobacter* and PSB liquid before transplanting. Irrigate with 5-7 cm of stagnant water. For top dressing, apply 40-48 kg N/ha at the tillering stage. For plant protection, control stemborer with light traps and spray Flubendiamide or Indoxacarb. Manage Leaf Blight with Streptocycline + Copper Oxychloride, and control Blast with Tricyclazole spray.

Soybean

Use recommended varieties: NRC-37, JS-335, KDS 726, Gujarat Soybean-2, and JS-9305. The ideal sowing time is from June to mid-July, with early sowing if conditions permit. Land preparation involves applying 10-15 tonnes of manure, followed by ploughing and harrowing. For spacing, use 45x7.5 cm or 30x5 cm, with a seed rate of 70-80 kg/ha. Fertilizer application should be 45:60:00 (N:P:K) kg/ha before sowing. Weeding can be done manually or with a Propaquizafop + Imazethapyr spray. For plant protection, control Bihar Hairy Caterpillar with Neem powder or pesticide spray, and manage Girdle Beetle with Novaluron or Quinalphos spray.

Pigeon Pea

For pigeon pea cultivation, early varieties include GT-1 and Vaishali, while late varieties are BDN-2 and IPCL-87119. The ideal sowing time is June-July, after rainfall. Land preparation involves applying 8-10 tonnes of manure, followed by ploughing and harrowing. For spacing, use 45-60 x 15-20 cm for early varieties and 90-120 x 25-30 cm for late varieties. Fertilizer application should be 25:50:00 (N:P:K) kg/ha.

OILSEED

Soybean is the main oilseed crop grown in Kharif season in Himachal Pradesh. Sowing of soybean should be completed from May end to mid-June using high yielding varieties Palam Soya, Harit Soya or Palam Hara Soya 1. For getting optimum population, 75-80 kg per ha seed is required. The row to row spacing of 45 cm and plant to plant spacing of 15 cm should be maintained with depth of seed not more than 3-4 cm. Farmers are advised to do seed treatment with carbendazim or Thiram @ 3-5 g/kg and with *Rhizobium* @ 20g /kg of seed before sowing. In case of aphid incidence apply thiametoxam @0.4 g/ L of water. Apply 20:60:40 dose of NPK kg/ha at sowing. Add sulphur @ 20 kg/ha for better oil content. Apply pendimethalin 750 g/ha or Oxadiazon 500 g/ha as pre-emergence for controlling many annual and broad leaf weeds. For post emergence weed control use Imazethapyr 100 g/ha or Imazethapyr 14 g/ha + Imazemox 14 g/ha in soybean.



Demonstration of soybean variety Palam Soya

PULSE CROPS

Blackgram

Among pulses, blackgram is the main pulse grown in Himachal Pradesh. Sowing of blackgram should be started by June end and completed in first fortnight of July using high yielding varieties UG 218, Him Mash 1 and PB 114 in low and mid hills and Palampur 93 in high hills (>1500 m amsl) using basal dose of NPK @ 20:40:20 at the time of sowing. Seed bio fortification with *Rhizobium* and PSB culture should be done before sowing for better plant stand and yield. For weed control, apply pre emergence herbicide Pendimethalin @ 1.5 kg /ha for weed control.



Seed Treatment with Biofertilizer



Demonstration of blackgram variety Him Mash 1 at farmers field

Rajmash (Kidney Bean)

High yielding varieties Jawala, Him-1, Kanchan, Triloki, Baspa of Kidney bean (Rajmash) can be sown in the month of May in high altitude areas and in the month of June in low lying areas. Weed control in kidney bean can be done using pre-emergence herbicide Pendimethalin @ 1.5 kg /ha. For management of anthracnose,

t/ha and incorporate during land preparation. Apply the following as basal dose during the final ploughing: Urea: 15 kg/ha, DAP: 130 kg/ha, MOP: 50 kg/ha. To control early-stage weed infestation, apply pre-emergence spray of Pendimethalin @ 1 kg a.i./ ha within 2-3 days after sowing (DAS). Recommended Varieties: Shalimar Moong-1 & Shalimar Moong-2

Rajmash (Kidney Bean)

Complete sowing in the first fortnight of May. Bush type: Seed rate: 60-70 kg/ha & Spacing: 30 × 10 cm. Pole type: Seed rate: 30-35 kg/ha & Spacing: 60 × 20 cm. Recommended Varieties: Bush Type: Shalimar Rajmash-1, Shalimar Rajmash-2 and Pole Type: Shalimar Rajmash-3, Shalimar Rajmash-4. All other practices are the same as those for moong.

Cowpea

Complete sowing in the first fortnight of May. Use a seed rate of 40 kg/ha. Maintain a spacing of 30 × 10 cm. Recommended Varieties: Shalimar Cowpea-1 & Shalimar Cowpea-2. All other practices are the same as those for moong.

Soybean

Complete sowing in the first fortnight of May. Use 40-50 kg seeds/ha. Maintain a spacing of 45 × 10 cm. Recommended Varieties: Shalimar Soybean-1 & Shalimar Soybean-2. All other practices are the same as those for moong.

DISEASES OF PULSES

Diseases of Rajmash, moong, cowpea: Symptoms: The infected plants show gradual wilting and drying. Yellowing and premature drying of leaves and stems are noticed. The basal portion of stem and root region becomes black. When the bark of infected root is peeled, black streaks are seen and vascular tissues show discolouration.

Leafspot: Symptoms: Leaves show light brown small spots. Shot holes are formed in due course. Lesions develop on petioles and stem

Angular leafspot: Initial symptoms appear on leaves as circular spots in between the veins which attains angular shape which are dark grey in colour and latter turn to reddish brown to dark brown.

Anthracnose: Most striking symptoms appear on immature pods although symptoms also appear on leaves, petioles and stem. Infection appears as brick red colour on the veins on lower surface of the leaves. On veins brownish black lesions appear on pods sunken cankerious spots with lighter or grey centers develop.

Rust: The rust pustules appear on all the green parts. On the leaves there are numerous small, orange brown pustules each surrounded by a light-yellow halo. Infected leaves wither and may fall. similar rust pustules as on leaves develop on stem.

Bean Common mosaic virus: Downward curling of foliage is diagnostic character in bean common mosaic virus. The leaves of infected plants become mottled, chlorotic, puckered and tend to droop, leaf size is reduced.

White Mold: Appearance of light brown water-soaked discolouration on the stem, leaves or pods and cottony growth on in collar region under humid conditions.

Pulses (especially pea, chickpea, and lentil) suffer White powdery fungal growth on leaves, leading to leaf yellowing and reduced yield. Carbendazim 50% WP @ 1.0 g per litre of water (apply at first sign of disease) or Tebuconazole 25% EC @ 1.0 ml per litre of water (apply preventively or when symptoms first appear) is advised. Blight (*Alternaria*, *Cercospora* spp.) affect chick pea, use of Mancozeb 75% WP @ 2.5-3.0 g per litre of water (apply when symptoms first appear) and copper Oxychloride 50% WP @ 2.5-3.0 g per litre of water (apply at early stages of disease) is advised. Aphids (*Aphis craccivora*) affect pulses especially chick pea, use Imidacloprid 17.8% SL @ 0.5 ml per litre of water or Thiamethoxam 25% WG @ 0.3 g per litre of water. For Podborer (*Helicoverpa armigera*) spray Spinosad 45 SC @ 0.3 ml/litre or Emamectin Benzoate 5 SG @ 0.5 g/litre at flowering and pod formation stages.

OILSEED CROPS

Farmers should select light sandy soil with good drainage for cultivation, and land preparation should begin in the first week of June. Recommended varieties for groundnut include K-1812, K-6, Dharni, Birsa Mungfali-3, Birsa Mungfali-4, and Birsa Bold. For niger, recommended varieties are Birsa Niger-1 and Birsa Niger-3. These seeds should be purchased in May and sowing should be completed before July 10. For sesame, suitable varieties include RT-351, Krishna, Kanke Safed, and Shekhar. For soybean, recommended varieties are Birsa Safed Soybean-2 and JS-2098, and for sunflower, DCS-107 is suggested.

Before sowing, seeds should be treated with carbendazim at 2 g/kg of seed and chlorpyrifos at 6 ml/kg of seed. At the time of sowing, fertilizer should be applied at the rate of 25:50:20:20 kg per hectare of N:P₂O₅:K₂O:S (Nitrogen: Phosphorus: Potash: Sulphur). For termite control (*Odontotermes* spp.), pre-sowing treatment should be done with Chlorpyrifos 20% EC at 2.0-2.5 ml per litre of water as a soil drench. For post-sowing treatment, apply Fipronil 0.3% GR (granular formulation) at 10-12 kg per hectare to control active termite populations. In case of rust (*Puccinia arachidis*), Propiconazole 25% EC should be used at the rate of 1 ml/ L of water as a foliar spray during early flowering or upon symptom appearance. To prevent damage caused by the Bihar hairy caterpillar, affected leaves should be plucked and buried in pits at the initial stage. If infestation worsens, Indoxacarb 15.8 SC should be sprayed at the rate of 10 ml per litre of water. For management of tikka disease or brown spots on leaves, Dithane M-45 should be sprayed at 1.25 kg /ha.

FRUIT CROPS

Mango

Mango varieties such as Amrapali, Dussehri, and Neelanchal Kesari should be irrigated at weekly intervals to ensure proper fruit growth and development. For quality fruit production, apply 500 grams of urea and 400 grams of muriate of potash per plant. During the fruit development stage, spray micronutrients on the leaves. After harvesting, provide adequate nutrition to the plants to restore their vitality. With the onset of the monsoon, sow green manure crops such as 20 kg of hemp or 10 kg of dhaincha per acre. Incorporate manure and compost into the soil 45 days after sowing. Pruning of mango trees should be carried out in the months of December and January.

October for Rabi crops. Recommended spacing is 60 cm × 20 cm or 60 cm × 30 cm, and sowing can be done by line sowing, seed-cum-fertilizer drill, or dibbling method. Fertilizer application varies with soil and irrigation conditions. For irrigated maize, a dose of 150:65:65 kg NPK per hectare along with 25 kg zinc sulphate is advised, while in dryland conditions, 100:50:25 kg NPK with 10 kg zinc sulphate is sufficient. Top dressing of urea at 45 days after sowing (DAS) improves yield. Application of micronutrients such as 19:19:19 at 5 to 10 g/litre of water and nano urea at 4 ml per litre is recommended during key stages like vegetative growth, tasseling, silking, and grain filling. Biofertilizers such as PSB, KSB at 500 g/ha, and Azospirillum at 500 g/ha for seed treatment further enhance nutrient uptake.

Intercropping systems such as maize with greengram, blackgram, cowpea, soybean, redgram (6:1), avare (8:1), and pigeonpea (4:2) are beneficial in increasing land-use efficiency. Irrigation should be provided every 8-10 days in red soils and every 10-12 days in black soils, with special attention during sowing, vegetative, and grain filling stages. Weed management includes one intercultivation at 30 DAS and hand weeding at 40 DAS. Atrazine 50% WP at 2-2.5 kg/ha is recommended as a pre-emergent herbicide, followed by post-emergent application of Tembotrione 42% SC at 287.5 ml/ha or Tambotrion 34.4% SC at 285 ml/ha. For disease management, Turicum leaf blight can be controlled with Mancozeb 75 WP at 2.5 g/litre, downy mildew with Metalaxyl + Mancozeb at 2 g/litre, and bacterial stalk rot with Copper hydroxide at 2 g/litre.

Seed treatment with Captan or Thiram at 4 g/kg of seed is also essential. Fall army worm is a major pest in maize and can be managed by seed treatment with Chlorantraniliprole at 0.4 ml/kg, spraying Emamectin Benzoate 5 SG at 0.2-0.3 g/litre, Chlorantraniliprole 18.5 SC at 0.2-0.3 ml/litre, and installing pheromone traps at 10 per hectare. Biological control with *Nomuraea rileyi* or *Beauveria bassiana* at 10 g/litre is also effective. Aphids can be managed with Azadirachtin 300 ppm at 5 ml/litre, while Spinetoram 11.7% SC at 0.5 ml/litre applied at 15-20 DAS helps manage early-stage armyworm attacks. In some cases, Diplodia ear rot has been observed at harvest.

OILSEED CROPS

Sunflower

Sunflower varieties suitable for cultivation include KBSH-44, KBSH-53, KBSH-78, KBSH-85, and KBSH-90. The recommended seed rate is 5 kg per hectare, and it can be sown during June-July, September-October, and December-January. Under irrigated conditions, apply fertilizers at the rate of 36:36:25 kg NPK/ha, and under rainfed conditions, apply 15:20:15 kg NPK/ha along with 4 kg of zinc and 6 kg of borax per hectare. Treat seeds with *Azotobacter* at 150 g/kg, Imidacloprid at 5 g/kg to control necrosis, and Metalaxyl at 6 g/kg for downy mildew. Maintain a spacing of 60 × 30 cm, and irrigate every 8-10 days in red soils and every 15-20 days in black soils. Apply a foliar spray of borax at 2 g/l during flower opening. For weed control, use Butachlor 50 EC or Alachlor 50 EC at 2.0 l/ha or Pendimethalin 38.7 EC at 1.8 l/ha. To manage diseases and insect pests, spray Imidacloprid 17.8 SL at 0.5 ml/l at 25 and 45 DAS for necrosis and sucking pests; Cypermethrin 35 EC at 1 ml/l for leaf-eating caterpillars; 5% NSKE or NPV (100 LE) for head borer. Practice thinning at 15-20 DAS and rotate with legumes for better productivity.

Groundnut

Recommended varieties are Dh 256 and Kadarilepaxi, sown during the rabi/summer season at a seed rate of 150 kg/ha. Fertilizer application includes 18 kg N, 46 kg P₂O₅, and 25 kg K₂O per hectare, along with 500 kg gypsum and 25 kg each of zinc and iron sulphate. Seeds are treated with 30 g of Captan 80 WP before sowing, and dibbling is done at a spacing of 30 x 10 cm. No intercrop is recommended. Foliar application of 20 g urea per litre of water is done at 30 and 60 days after sowing. Irrigation is crucial at flowering and pod development stages.

Inter-cultivation is carried out at 15, 30, and 40 days after sowing. Weed management includes pre-emergent application of 1.3 litres of Pendimethalin 30 EC. Major diseases include leaf spot, managed by spraying 1 g Carbendazim 50 WP or 1 ml Hexaconazole, and collar rot, managed by seed treatment with 3 g of Carbendazim 25% + Mancozeb 25%. Pest control includes spraying 2 ml of Profenofos 50 EC or 1 ml of Monocrotophos 36 SL for leaf miner, and 2 ml of Quinolophos 25 EC for Spodoptera caterpillar. Application of 2 quintals of neem cake before sowing is also advised.

PULSE CROPS

Redgram

Recommended varieties of redgram include TS-3R, GRG-811, and GRG-152. Use 10-12.5 kg of seed per hectare and sow during June-July. Apply fertilizers at 25:50:75 or 25:50:125 kg NPK per hectare in red soils and 25:50:0 in black soils. Treat seeds with *Rhizobium* 500 g, PSB 500-1250 g, PGPR 500 g, Trichoderma 10 g/kg, and soak in 2% calcium chloride solution followed by shade drying for 7 hours. Use a spacing of 90 x 30 cm under normal planting, 120 x 20 cm for wider rows, and 45 x 10 cm for compact planting. Redgram can be intercropped with bajra, little millet, greengram, soybean, or blackgram. Apply pulse booster or pulse magic during flowering and pod-filling stages.

For weed management, use Pendimethalin at 3.25 L/ha as pre-emergence and Imazethapyr at 1 L/ha at 3 weeks after sowing. For disease management, use *Trichoderma*-treated seeds for wilt, Oxydemeton-methyl at 1.5 ml/L for sterility mosaic, and Metalaxyl + Mancozeb at 4 g/kg seed for *phytophthora* blight. To control pod borers, use Emamectin benzoate at 0.3 g/L or Coragen at 0.3 ml/L. For *Helicoverpa*, apply Profenophos at 2 ml/L, NSKE at 3 ml/L, or Chlorantraniliprole at 0.15 ml/L. Practice nipping at 45-50, 55-60, and 75-80 days after sowing to promote branching.

Greengram

Recommended varieties for greengram include DGGV-2, BGS-9, Pusa Baisakhi, and DGGV-1. The seed rate varies from 7.5 to 15 kg/ha, and sowing is done during June-July. Apply fertilizers at 12.5:25:20 or 25:50:0 kg N:P:K/ha as a basal dose at sowing. Treat seeds with *Rhizobium* 500 g and PSB 1250 g. Maintain spacing of 30 x 10 cm or 37.5 x 10 cm. It can be intercropped with redgram or sugarcane. During flowering and pod formation, spray pulse booster or pulse magic at 8-10 g/l.

For weed control, apply Pendimethalin at 3.25 l/ha immediately after sowing.

90, 120, 150, or 180 cm) and the pit method (spacing of 90 cm x 45 cm). Intercropping with onion (1:2 ratio) and soybean (1:1 ratio) is recommended during the early stages of crop growth.

For irrigation, follow a stage-wise schedule: every 7 days during germination (8-35 days), every 10 days during tillering (36-100 days), every 7 days during the grand growth stage (101-270 days), and every 15 days during ripening (271-365 days). Weed management involves manual or mechanical weeding at 30, 60, and 90 days after planting, along with chemical control. Pre-emergence herbicide application of Atrazine 50 WP (1 kg in 300 liters of water/ha) should be done 2-3 days after planting, while post-emergence control at 60 days after planting includes spraying 2,4-D sodium salt 80 WP (1 kg in 300 liters of water) or Mesotrione 40 SC (140 ml in 200 liters of water/acre).

For pest and disease management, rhizome weevil infestation can be controlled by treating rhizomes with 20% Neem Seed Kernel Extract. To control leaf-eating caterpillars, apply 2.5 ml/liter of Chlorpyrifos, and for thrips, use 0.3 ml/liter of Imidacloprid or 1.5 ml of Dimethoate per liter. Leaf spot disease can be managed with 1 ml/liter of Propiconazole. For nematode control, apply 12 grams of *Pseudomonas*, 12 grams of *Pachelomyces*, and 500 grams of Neem Cake. Early shoot borer management includes earthing up at 6 and 10 weeks, removing infested shoots, and releasing *Trichogramma parasitoids* (50,000/ha) at 7-day intervals for five times.

VEGETABLES CROPS

Onion

Commonly grown onion varieties include Arka Kalyan, Arka Niketan, Arka Pragati, Bhima Super, and Bhima Red, while in the Gadag region, Bhima Super is the most preferred variety. The recommended sowing seasons across both regions are June-July, September-October, and January-February, depending on the crop type. For transplanting, 8-12 kg of seed per hectare is required, while seed drill sowing needs 20-25 kg/ha. In bulb planting, approximately 875 kg of bulbs per hectare is used. A general seed rate of 5-6 kg/ha is recommended when using Bhima Super or Arka Kalyan varieties. Before sowing, seed treatment with Captan 80 WP at 2 g/kg of seed helps control fungal diseases, and Imidacloprid 60 FS at 15 ml/kg of seed is essential for protection against sucking pests. The recommended fertilizer dose is 30 tonnes of FYM per hectare, along with 125:75:125 kg/ha of N:P:K fertilizers. In Belagavi, sowing methods include seed drill sowing, transplanting, and bulb planting, whereas in Gadag, broadcasting is commonly practiced. For transplanting, a spacing of 15 cm between rows and 10 cm between plants is ideal. In general, criss-cross or line sowing with a spacing of 45 × 30 cm is also recommended. In the Gadag region, farmers commonly intercrop onions with chilli, coriander, or wheat. To enhance plant growth, foliar sprays of Arka Vegetable Special or 19:19:19 formulation at 3-5 g/litre of water are beneficial. Additionally, spraying Miraculin at 2000 ppm (2 ml/litre) 50 days after sowing helps improve yield. Irrigation should be provided once every 4-5 days, based on soil and weather conditions, and must be stopped 15 days before harvesting. While onion in Gadag is mostly rainfed, it requires supplemental irrigation during dry spells.

OILSEED CROPS

Soybean

Use improved high yielding soybean varieties such as JS 20-69, RVS 2002-4, JS 20-98, JS 20-116, Indore Soya 142, NRC 150, NRC 152, JS 2172, RVSM 2011-35 and Indore Soya 130. To eradicate the pest propagules, including the pupae of major pests and the spore structures of major pathogens that remain on farms, such as on threshing floors and in the soil, it is essential to minimize the initial pest/pathogen load. Sanitation of threshing floors, deep ploughing of fields, and cleaning of bunds are necessary. Farmers should also collect all crop debris and bury it in a compost pit.

Seed materials should be carefully processed to remove physically damaged, discolored, or undersized seeds. Only clean, healthy, pest- and disease-free material should be stored as seed. Also treat the seeds with Carbendazim 25% + Mancozeb 50% WS @ 3 g/kg seed + biological fungicide *Trichoderma viride* @ 8-10 g/kg seed, *Bradyrhizobium* culture and Phosphate Solubilizing Bacteria @ 10 g/kg seed each to get higher yields. For the control of stem fly in soybean, treat seeds with the Thiamethoxam 30 FS @ 10 ml/kg seed before sowing.

Ensure that soybean is sown on raised bed and Broad Bed and Furrow methods, to conserve moisture. For early maturing and short-statured varieties, maintain 30 cm row spacing. For long-duration and tall varieties, maintain 45 cm spacing between rows and 10 cm between plants. Apply fertilizers based on soil test recommendations. Generally, N: P₂O₅: K₂O at 25:60:40 kg/ha along with 20 kg/ha bentonite sulphur is recommended.

VEGETABLE CROPS

Nursery management: Prepare nursery beds with deep summer ploughing. To protect the nursery from hot winds in June, sow dhaincha seeds along the borders. Always prepare raised nursery during monsoon season and sowing should be done in line to avoid mortality by damping off disease. Soil application of *Trichoderma harzianum* culture @ 100 gm per bed during preparation of nursery bed is very helpful to escape various fungal diseases. Prepare tomato, brinjal, and chili nursery under 50% green shade net, which help reduce vector-borne diseases and thereby ensure the production of healthy seedlings. To protect the nursery from direct sunlight, cover it with a net at a height of 6.5 feet. Sow early-maturing cauliflower on raised beds @ 600-700 g/ha. Treat seeds with Thiram 75 % DS @ 2.0-2.5 g/kg of seed before sowing. Irrigate moderately in the morning or evening to maintain soil moisture.

Field Management: The current weather is favorable for onion cultivation. Before sowing, treat the seeds with Captan 75 WS @ 2.5 g/kg seed. In transplanted onion, dip the onion roots in a solution of Carbendazim 12% + Mancozeb 63% @ 2.5 g/L of water prior to transplanting. Raise the onion nursery with high-yielding varieties such as Agrifound Dark Red, Pusa Ridhi, N-53, Bhima Super, Bhima Raj, and Bhima Dark Red @ 12-15 kg seeds/ha. Considering the current temperature, farmers can begin garlic sowing. Ensure that the soil has adequate moisture before sowing. Recommended improved varieties to be sown such as G-1, G-41, G-50, and G-282.

During field preparation, apply the recommended dose of farmyard manure along with phosphorus fertilizer. Top dressing of nitrogen @ 20-25 kg/ha for cucurbit crops and 30-40 kg/ha for tomato, brinjal, and chili to be done for getting higher yield. For better fruit setting in tomato, brinjal, and chili, spray



The Kharif season, stretching from June to October, is a crucial agricultural period for the state of Maharashtra, significantly influencing the livelihoods of millions of farmers. Characterized by the onset of the southwest monsoon, this season supports the cultivation of key crops such as rice, cotton, soybeans, pulses, and millets. However, climate variability, erratic rainfall, and changing weather patterns pose continuous challenges to sustainable agricultural production. In the light of these challenges, the Kharif 2025 Agro-Advisory aims to provide timely, region-specific, and science-based recommendations to empower farmers with the knowledge needed to make informed decisions.

REGIONAL ADVISORY

Maharashtra is divided into several agroclimatic zones, each with unique characteristics influencing crop selection, sowing windows, and input management. The following region-wise advisories aim to support farmers with tailored recommendations for improved productivity and resilience.

VIDARBHA REGION

The Vidarbha region of Maharashtra is composed of Bhandara, Chandrapur, Gondia, Gadchiroli, Nagpur, Amravati, Akola, Yavatmal, Buldhana, Wardha, Washim and possesses state agricultural university viz., Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola.

Soil Testing & Preparation (All Crops)

Soil testing is crucial for effective fertilizer management, optimizing input use, and maintaining soil fertility. For proper soil sampling, the recommended depths are 15-20 cm for crops like sorghum, rice, groundnut, and wheat; 30 cm for cotton, sugarcane, and banana; and 30 cm for fruit trees, sampled from a distance of 1-1.5 feet from the trunk. Additionally, applying decomposers to crop residues like straw and leaves enhances organic matter content, improves soil structure, and boosts water retention, contributing to sustainable agricultural practices.

COMMERCIAL CROPS

Cotton

The cotton crop is sown during May-June. Land preparation includes deep ploughing every three years for rainfed areas and annually for irrigated ones, followed by two harrowings and one pre-sowing harrowing. FYM or compost is applied at 12-15 cartloads for rainfed and 20-25 cartloads or 5 t/ha for irrigated fields, with 2.5 t/ha vermicompost if available. Seeds are treated with Imidacloprid or Thiomethoxam (7.5 g/kg), Thiram or Captan (3 g/kg), and biofertilizers *Azotobacter* (25 g/kg) and PSB (20 g/kg). Sowing times vary: 20-30 May for irrigated, early June for dry sowing, and 15-30 June post-monsoon. For Bt/Hybrid (irrigated), spacing is 120x90 cm with a seed rate of 2-2.5 kg/ha, while Deshi improved (rainfed) is sown at 60x15 or 60x30 cm with 9-15 kg/ha. Intercropping with green gram, black gram, or soybean (1:1) or with sorghum and pigeon pea (6:1:2:1) is recommended. Gap filling and thinning are done within 3-5 days after emergence. Irrigated Bt requires 40:60:60 NPK, while Hybrids need 35:50:50 NPK, both applied in two splits. Rainfed varieties receive 20-30:20-30:20-30 NPK as a basal dose using the ring method. Delayed application reduces yield by 10-40%. Foliar sprays of 2% urea (flowering) and 2% DAP (boll stage) are advised. Drip fertigation involves staged application: 10% at sowing, 20% at 20 DAS, 25% each at 40 and 60 DAS, and 20% at 80 DAS, with phosphorus applied as basal. Weed control is crucial for the first 60 days, requiring 3-4 hoeings and 2-3 weedings. Moisture conservation is done by opening furrows at 40-60 DAS. Rainfed cotton benefits from 2-3 protective irrigations during flowering and bolling.



High Density planting in cotton

Common pests like Jassid, Thrips, Aphids, and Mites are managed with Dimethoate 30 EC (10 ml/10 L) or Acetamiprid 20 SP (1.5 g/10 L), while whiteflies are controlled using yellow sticky traps (5 per acre) and Dimethoate 30 EC (33 ml/10 L). Bollworms are managed with Quinalphos, Spinosad, or Cypermethrin, and pink bollworm with pheromone traps, Neemark 300 ppm, deep ploughing, and heat treatment. Thirty yellow sticky traps and 2-3 light or pheromone traps per gin are suggested. Biopesticides like Neem 5%, *Verticillium lecanii*, and *Beauveria bassiana* are recommended in humid weather.

Plant Protection: Bacterial leaf blight is managed by treating seeds with Carboxin (1 g) and Thiram (3 g/kg), grey mildew with wettable sulphur (25 g/10 L) or Carbendazim (10 g/10 L), Myrothecium with Copper oxychloride (0.3%), Carbendazim (0.1%), or Mancozeb (0.25%), and *Alternaria* with seed treatment and Mancozeb (0.2%) sprays every 20 days. Para wilt is managed with a drench of 25 g Copper oxychloride and 150 g urea in 10 L of water.

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Special recommendations include intercropping AKH-081 with green or black gram, spacing PKV DH-1 at 60x45 cm with 50:25:25 NPK (rainfed), and applying 2% urea (flowering) and 2% DAP (bolling) foliar sprays. For AKH-081 in shallow soils, 5 t/ha FYM and 25:12.5 N:P are advised. For pre-monsoon yield boost, 2.5 t FYM and 75:37.5:37.5 NPK/ha are applied.

Sugarcane

Sugarcane is planted during the Kharif season (June-July) with a duration of 12-15 months. It thrives best in well-drained deep loam soils with good water-holding capacity, avoiding saline or alkaline conditions. The climate should be warm and humid. Land preparation involves deep ploughing (1-2 times) followed by two cross harrowings, stubble and weed removal, and proper field leveling with drainage. Recommended varieties include CO-86032, CO-8014, CO-94012, VSI-434, and CO-M-0265, especially early-maturing ones. Healthy 9-10 month-old canes are selected for setts, taking the top two-thirds portion and avoiding flowering canes. The seed rate is 45,000-50,000 three-budded setts per hectare, approximately 6-8 tonnes. Before planting, setts are treated with 0.25% Bavistin or Carbendazim mixed with a 2.5% urea solution for 10-15 minutes.

Spacing varies: 90 cm for rainfed, 120 cm for irrigated or paired row trench planting, and 60 cm for High-Density Planting Systems (HDPS). Planting is done in furrows or trenches, with buds facing upward, covered lightly with soil. The fertilizer dose is 250:115:115 NPK kg/ha for irrigated crops and 200:90:90 NPK kg/ha for rainfed crops. At planting, 1/3 nitrogen (N) along with full phosphorus (P) and potassium (K) is applied, while the remaining N is split and applied at 45 and 90 days after planting (DAP). Fertilizers are placed in bands 8-10 cm deep and 10-15 cm away from the rows. Additionally, 25-30 t/ha of FYM or compost is applied before the final harrowing. Biofertilizers such as Azotobacter and PSB (2 kg/ha each) mixed with 100 kg of compost or FYM are added during planting. Intercropping with green gram, black gram, soybean, or groundnut is recommended during early stages. Light earthing up is done at 45-60 DAP, followed by full earthing up at 90-120 DAP. Trash mulching with cane trash after 60 DAP helps conserve moisture and control weeds.

For pest control, early shoot borer is managed by removing affected shoots, releasing *Trichogramma* (@ 50,000/ha/week), or applying Chlorpyrifos 20 EC (5 ml/L). Internode borer is controlled with Cartap hydrochloride 4G (10 kg/ha) or *Trichogramma* release. White grub management includes applying Phorate 10G (25 kg/ha) in moist soil. Red rot is prevented by planting resistant varieties, removing infected clumps, and treating setts with Carbendazim 0.1%. Smut is controlled by using disease-free setts, hot water treatment (50°C for 30 minutes), or fungicide dips. Wilt is managed through resistant varieties and crop rotation. For growth enhancement, 2% urea or 2% DAP is applied during tillering and grand growth stages if the crop appears pale or yellow. These practices ensure healthy growth and optimum yields.

PULSES

Soybean

Sowing of soybean is recommended between 20th June and 5th July, with a seed rate of 65 kg/ha and a spacing of 45 cm. Preferred varieties include MAUS-162, MAUS-612, PKV Amba, and PKV Yellow Gold. Seeds should be treated with a fungicide-insecticide combination like Azoxystrobin (10 ml/kg) and inoculated with Rhizobium and PSM (5 g/kg). For weed control, apply Pendimethalin 30% EC (3.25 L/ha) or Flumioxazin 50SC (250 ml/ha) as a pre-emergence herbicide, followed by

PULSE CROPS

Soybean

The own seed of Soybean should be used for next Kharif season after testing its germination to save money as well as it will help to face the situation of seed shortage in the market. Seed germination should be tested before sowing and good quality seed with more than 70 % germination should be used for sowing only. Improved varieties like MAUS-162, MAUS-612, MAUS-725, Phule Durva, Phule Kimaya, PDKV Yellow Gold, PDKV Amba should be used for sowing. Soybean seed should be treated with Thiram @ 3 g/kg or *Trichoderma* @ 4 g/kg of seed and inoculated with *Rhizobium* and PSB @ 100 ml/10 kg of seed before sowing. Sowing of soybean should be avoided in the fields where soybean seed production was taken in the last summer season. Follow IPM package for the control of pests and diseases in soybean as give below:

- Sowing should be completed before 15th July to avoid pest infestation.
- Seed treatment of Thiram @ 2-3 g per kg or *Trichoderma* @ 5 g per kg of seed.
- Seed treatment with Thiamethoxam 30 % @ 10 ml per kg of seed for management of stem fly.
- Use border crop of castor and sunflower as a trap crop for *Spodoptera*.
- Install 16 to 20 bird perches per ha.
- For management of *Spodoptera* spray 5 % NSKE or *Beauveria bassiana* or *Numeria releyi* @ 40 g or *Bacillus thuringiensis* @ 20 g per 10 L of water.

Pigeon Pea

The Sowing time for early varieties is first fortnight of June and medium/late: Second fortnight of June to first fortnight of July. Seed rate for early: 20-25kg/ ha spacing: 45X60cm R-R and Medium to Late: 15-20kg/ ha spacing: 60-75cm R-R. Use wilt and sterility mosaic resistant medium to early duration varieties like BDN-711, BDN-716, BDN-2013-41 (Godavari), PKV Tara, Phule Rajeswari, etc to ensure higher yield in adverse climatic conditions. Treat the seeds with biofertilizers and apply recommended doses of fertilizers as per recommendations of concerned universities. Ensure arrangement of one or two protective irrigation wherever possible. Intercropping of Pigeon pea + Soybean with 1:2 or 2:4 ratio should be taken for higher yield and monetary returns. Install 5 pheromone traps per hectare for monitoring and spray 5 % NSKE at flowering stage for the control of *Helicoverpa*.



Wilt in Pigeon pea

Green Gram and Black Gram

Sowing of green gram should be completed with first onset at minimum 80 mm of rainfall. Improved varieties of green gram like BPMR-145, BM-2003-02, Phule Chetak, etc and AKU-10-1 (Black Gold), AKU-15, etc of black gram should be used for

Pokkah boing Disease spray the crop with Mancozeb 75 WP @ 30 g or Carbendazim 50 WP @ 10 g in 10 litre of water. For the management of Rust and Leaf spot diseases spray the crop with Mancozeb 75 WP @ 30 g or Tebuconazole 25 EC @ 10 ml in 10 litre water.

PULSE CROPS

Pigeon pea

Select proper sowing windows based on maturity groups. Early varieties can be sown in First fortnight of June and Medium/Late varieties in Second fortnight of June. Ensure flowering avoids low sunlight/cloudy weather to prevent poor pod formation. Use well-drained black cotton soil (pH 7.0-8.5). Adopt deep ploughing or BBF/Ridge & Furrow systems to improve drainage and aeration. Raised bed planting and appropriate row spacing (e.g., 4-5 ft) for better management and intercropping potential. Use quality seeds with appropriate seeding rates: Early: 20-25 kg/ha. Medium/Late: 15-20 kg/ha. Treat seeds with *Trichoderma viride*, Thiram + Carbendazim, and Rhizobium + PSB for disease prevention and better nitrogen fixation. Fertilization: Apply 25-30 kg N, 40-50 kg P₂O₅, 30 kg K₂O/ha at sowing. Incorporate Sulphur, Zinc, and Iron if soils are deficient. Follow soil test-based recommendations for precision nutrition. Irrigation & Drainage: Though drought-tolerant, apply 3 key irrigations during branching, flowering, and podding stages. Use ridge planting in poorly drained soils to avoid waterlogging. For weed Management: Pendimethalin (0.75-1 kg a.i./ha) as pre-emergence to suppress early weeds. Conduct two mechanical weedings before flowering (20-25 DAS & 45-50 DAS). Cropping and Intercropping Systems: Utilize inter-row space with short-duration crops: mung, urd, cowpea, , and recommended intercropping arwe Pigeon pea-Urd-Wheat, Pigeon pea-Sugarcane, Pigeon pea (early)-Potato-Urd bean.

Pest & Disease Management: Use resistant varieties (e.g., Asha, BSMR-853, JKM-189). Follow integrated pest management (IPM): pheromone traps, Neem products, timely chemical sprays (Emamectin, Indoxacarb).

Soybean

Soybean should be sown by the row-to-row spacing is 45 cm, and plant-to-plant spacing is 7 to 10 cm. A seed rate of 65 to 75 kg per hectare is sufficient, depending on the seed size and germination percentage.

Soybean should be treated with Thiram @ 3g/kg or *Tricoderma* @ 4 g/kg of seed inoculum like *Rhizobium* and PSB @ 100ml/10kg of seed before sowing. For higher production use new released high yielding varieties like Phule Sangam (KDS 726), Phule Kimaya (KDS 753), Phule Durva (KDS 992. Apply NPK @ 50:75:40 Kg/ha at the time of sowing. Use Broad bed furrow planter (BBF) for sowing. It will be beneficial during less rainfall situation and excessive rainfall. For chemical weed control, Pendimethalin at 1.0 kg active ingredient per hectare can be applied as a pre-emergence herbicide immediately after sowing. Imazethapyr at 75 grams active ingredient per hectare can be applied as a post-emergence herbicide at 15-20 days after sowing to control broadleaf and grassy weeds.

follow with hand weeding at 20-25 DAS. Monitor for sucking pests (jassids, aphids, whiteflies); control using imidacloprid 0.3 ml/l or thiamethoxam 0.25 g/l.

Further, install 5 pheromone traps per hectare to monitor bollworms. If needed spray emamectin benzoate or flubendiamide as per threshold levels. Manage leaf spots and blights with mancozeb or copper oxychloride sprays as needed. Apply top dressing of nitrogen at 30 and 60 DAS for vigorous growth. Use plant growth regulators like NAA to minimize flower and boll drop, if observed. Harvest mature, dry bolls; avoid harvesting during or immediately after rain to prevent boll rotting.

PULSE CROPS

Soybean

The crop must be sown before 1st week of July, after good rain, with a spacing of 45×57cm; depth <4?cm. the seed must be treated with Carboxin + Thiram 30g/10kg + Thiamethoxam 100ml/10kg. Give a hoeing at 15-20 DAS. Apply water at branching (30-35 DAS) & flowering (40-45 DAS).

Pigeon pea

Recommended varieties for sowing include Phule Trupti, BDN-711, and Renuka, with planting done before the first week of July. Suggested spacing is 60×20 cm, 90×20 cm, or 180×30 cm, depending on field conditions. Treat seeds with Trichoderma (5 g/kg) or a combination of Carboxin and Thiram (4 g/kg), along with Rhizobium and PSB at 25 g/kg each. Apply fertilizers at a rate of 25:50 kg N:P per hectare. The seed rate varies-12-15 kg/ha for normal spacing and 5-6 kg/ha for wider spacing. Intercropping is recommended with bajra, soybean, sorghum, cotton, moong, or urad for better land utilization. Irrigate at 30-35 days after sowing (DAS), 60-70 DAS, and during the pod-filling stage. Perform one hoeing at 15-20 DAS followed by hand weeding for effective weed control.

Black gram

The recommended seed rate is 15-20 kg/ha with a spacing of 30×10 cm. Seeds should be treated with Trichoderma at 5 g/kg along with Rhizobium and PSB at 25 g/kg each. Apply fertilizers at a rate of 20:40 kg N:P per hectare. Intercropping with pigeon pea and other kharif crops is advised for better resource utilization. Irrigation should be provided at critical stages such as flowering and pod filling to ensure healthy growth and yield.

FRUIT & VEGETABLE CROPS

Pomegranate (Mrig Bahar)

Pomegranate cultivation during Mrig Bahar requires pruning to remove diseased branches, followed by an application of 1% Bordeaux mix. Pre-flowering fertilization includes 700 g urea, 1600 g SSP, and 400 g MOP per tree. Dusting with 4% copper dust or bleaching powder helps in disease prevention. Regular pest control with Azadirachtin or alternate insecticides every 15 days is crucial. At the fruiting stage, apply 500 g DAP + 100 g MOP per tree at the lemon size stage, and 200 g 19:19:19 + 100 g MOP at the guava size stage.

60 kg N, 30 kg P_2O_5 , and 30 kg K_2O per hectare for optimal crop nutrition. Millets can be intercropped with soybean. Conduct three hoeings with a hand hoe during early growth stages for effective weed suppression. Spray Mancozeb (0.2%) at early crop stages if Blast, Brown Spot, or Rust symptoms appear. Rogue out diseased plants to manage spread and for armyworms, cutworms, and leaf-scraping beetles, apply Thiamethoxam @ 5 g/15 litres of water for effective protection.

PULSE CROPS

Blackgram

Conduct deep summer ploughing to expose and destroy the eggs, larvae, and pupae of major pests like pod borer, caterpillars, and weevils. Mix FYM/compost into the soil 10-15 days before sowing. Ensure the field is free from weeds for optimal crop growth. Use high-yielding varieties like PU 31 and LBG-20 (Teja). PU 31 is tolerant to Yellow Mosaic Virus (YMV) and well-suited for Kharif cultivation, while LBG-



Varietal demonstration in blackgram

20 performs well in all seasons. Use a seed rate of 10-15 kg/ha (or 2.5-3.75 kg/sangam). Sowing can be done by broadcasting or dibbling at a spacing of 15 cm × 30 cm. If required, apply lime in the furrows at sowing, preferably every alternate year, based on soil test recommendations to manage soil acidity. Treat seeds with Carbendazim @ 2g/kg of seed before sowing to protect against seed-borne diseases and promote healthy germination. Apply 50 kg potash/ha to the crop raised with closer spacing (20X10cm) to reduce stem fly and pod borer incidence. Treat the seeds with Carbendazim 50WP or Thiram @ 2g/kg of seeds or with Trichoderma viride @ 4g/kg of seeds before sowing. Install yellow sticky traps @ 10 traps/ha in the field to manage aphids. Spray Dimethoate 30EC @ 1ml/litre of water or Imidacloprid 17.8SL @ 0.5ml/litre of water for managing sucking pests.

OILSEED CROPS

Groundnut

Perform one ploughing with a soil-turning plough followed by two harrowings to achieve fine tilth up to 12-18 cm. Sow from 1st week of May to 1st week of June for optimal growth. Use ICGS 76 (semi-spreading, matures in 120-125 days) and JL 27 varieties. Treat groundnut seeds with Dithane M-45 or Thiram @ 3 g/kg to protect from seed and soil-borne diseases. For both soybean and groundnut, seed treatment is crucial under dry conditions. Keep the groundnut field weed-free for up to 50 days after sowing, as weed pressure is high due to monsoon rains. Timely inter-cultivation and weeding are essential for higher yields. Perform earthing up at 30-35 days after sowing in groundnut to promote pegging and pod development. Promote groundnut and soybean in foothill areas with limited irrigation, where rice cannot be grown, as climate-resilient and economically viable alternatives.

spacing of 60 cm x 30 cm. Intercropping with Dwarf French bean (var. Pant Anupam) or soybean is recommended for better resource utilization and additional income. Treat maize seeds before sowing with *Trichoderma harzianum* @ 5 g/kg seed or *Azospirillum/Phosphobacteria* @ 20 g/kg seed to improve germination and disease resistance. Carry out manual weeding and earthing up in early-sown maize to prevent lodging due to heavy winds. If any Fall Army Worm (FAW) infestation is observed, apply dry soil in the whorl of infected plants or spray neem oil (1500 ppm) as a first-line defense. For biological control, release *Trichogramma pretiosum* @ 50,000 eggs/ha and spray *Metarhizium anisopliae* @ 5 g/l of water into the leaf whorls 15-25 days after planting for effective FAW management.

MILLETS

Finger millet is a major dual-purpose crop grown during the kharif season for both grain and fodder. It is highly drought-tolerant and is usually sown with the onset of the monsoon. Farmers are advised to select suitable drought-tolerant varieties such as VL 376 (105 days), VL 379 (early maturity in 100 days), and VL 380 (mid-late maturity in 120 days). Transplanting of seedlings should be done 21 to 30 days after sowing in the nursery. Delayed transplanting beyond this period may significantly reduce tillering, resulting in lower yields. To ensure an optimum plant population, use 4-5 kg of seeds per hectare.

VEGETABLE CROPS

Farmers are advised to prepare the vegetable nursery for crops such as cucumber, bitter gourd, ridge gourd, bottle gourd, chilli, brinjal, and okra on raised bamboo structures, covering them with polythene to protect seedlings from heavy rain and hailstorms. Planting should be done in ridges and furrows with proper drainage channels to prevent waterlogging. It is essential to add farmyard manure (FYM) or vermi-



In-situ moisture conservation in cabbage

compost in the nursery for healthy seedling growth. To conserve soil moisture, prevent soil erosion, and control weeds, mulching with locally available materials like paddy straw is recommended. Farmers can also make their own compost using the 18-day compost method, pit method, or by producing vermicompost to meet their manure requirements. For pest management, it is advised to make biopesticides using locally available materials such as leaves, garlic, and tobacco. For crops under protected cultivation, including high-value crops, frequent irrigation should be provided at 10-15 day intervals, based on the weather conditions.

SPICE CROPS

Ginger and Turmeric

Farmers are advised to complete the sowing of ginger (variety Nadia) and turmeric (varieties Megha Turmeric 1 or Lakadong) without delay. Use a seed rate of 15-20

quintals per hectare and maintain a spacing of 30 cm × 30 cm. Before sowing, treat rhizome seeds by dipping them in a solution of *Trichoderma viride* @ 4 g/litre of water for 5 minutes or *Trichoderma harzianum* @ 10 ml/litre to prevent rhizome rot. For already sown crops, carry out proper weeding and earthing up to promote healthy growth. Ensure planting in ridges and furrows with good drainage. Farmers may also opt for intercropping with crops like soybean, rice bean, French bean, finger millet, or maize for better resource use and income diversification. To conserve soil moisture and control weeds, mulch the beds using dry paddy straw or green leaves. For effective disease management, especially soft rot, spray *Trichoderma viride* or *Trichoderma harzianum* @ 10 ml/litre or neem oil (10,000 ppm) @ 2 ml/litre as a preventive measure.

LIVESTOCK

Piggery

Farmers are recommended to select pig breeds like Lumsniang or Hampshire cross; supplement mineral mixture (50-60g/day) and calcium for lactating/pregnant sows; use local feed alternatives like banana pseudo stem, colocasia, and sweet potatoes. Ensure adequate clean drinking water and proper feed. Deworm pigs with Albendazole/Febendazole (7.5mg/kg body weight every 3 months); vaccinate against Swine Fever; regularly provide 20g of mineral mixture per pig with feed; and isolate sick pigs in a quarantine shed until fully recovered. Maintain dry, clean pig sheds with proper roofing and walls; avoid waterlogging; disinfect farm entrances and exits with 1% formaldehyde or 2% NaOH; and ensure footwear and equipment are disinfected. Quarantine newly purchased pigs for 4 weeks before introduction. Implement strict measures for African Swine Fever (ASF) by quarantining infected farms, prohibiting the movement of animals and personnel between farms, and ensuring the proper disposal of carcasses (deep burial with lime/bleaching powder). Avoid swill feeding unless properly boiled; maintain proper records of animal health and visitors; and prohibit animal movement between farms to reduce disease risk. Report suspected ASF cases to veterinary authorities immediately.

Poultry

Farmers should maintain clean, ventilated poultry sheds with proper space (1 ½ sqft/bird); avoid overcrowding and heat stress in birds. Regularly change litter materials to prevent coccidiosis and maintain hygiene. Provide clean drinking water 3-4 times a day, occasionally adding electrolytes. Ensure the water is free from contamination and provide proper feed to maintain bird health. Backyard poultry



Vaccination in poultry

breeds like Vanaraja, Shrinidhi, and Kuroiler can be reared in low-cost housing systems. Ensure the houses are clean, dry, and properly ventilated to prevent infections and diseases. Administer anticoccidial drugs in drinking water for 3-5 days

local short duration var. Idaw, tai, Buhsakei are recommended for upland/Jhum cultivation with rich alluvial soil

Pest & Disease Management: To effectively control Gundhi bugs, deploy 100 rotten crab funnel traps per hectare during the critical milking stage of the crop. For organic pest management, utilize neem-based biopesticides, release *Trichogramma* egg parasitoids, and install yellow sticky traps to monitor and reduce pest populations. Additionally, conserve natural predators such as spiders, dragonflies, and ladybugs by minimizing chemical pesticide use and maintaining ecological balance in the field. Adopt SRI method in small areas for enhanced yields, conduct regular field monitoring for pests and nutrient deficiencies and maintain proper water management and spacing as per cultivation method.

OILSEED CROPS

Soyabean

Varietal Selection: RCS1-1, RCS1-9, RCS1-10, JS335, MAUS 71 is suitable option for Kharif cultivation in Mizoram.

Irrigation: Despite it is a rainfed crop in kharif, Supplemental irrigation is required depending on rainfall patterns. Flowering and pod filling stages are the critical stages of irrigation.

Relay cropping: Soybean can be grown as a relay crop with maize, planting between the rows of the maize crop is recommended.

Integrated nutrient management: Soybean being a leguminous crop does not require high dose of nitrogen, however 20 kg/ha is sufficient for healthy crop stand. A dose of 60 kg P_2O_5 (SSP) and 30kg K_2O (MOP) are recommended. FYM @ 5t/ha should also supplement the chemical fertilizers.

Integrated disease management: Seed treatment with Thiram at the rate of 4.5g per kg of seed will control fungal diseases like Leaf Blight, Leaf Spot, Seedling Rot, Frog eye spot and Rust.

Integrated pest management: Spray application of Nuvacron 1.25ml/L or Dimethoate (0.04%) can be used for effective control of leaf folder, stem fly and semilooper like pests. Seed treatment with Imidachloprid @7g/kg of seed reduces the problem of Leaf folder

Groundnut

For optimal groundnut production, farmers should complete sowing between 15th May and first week of June, using high-yielding varieties like ICGS 76, ICGV 86590, Girnar 4, TS 67 Before sowing, treat seeds with Bavistin (5g/kg seed) for disease prevention and apply Rhizobium + PSB culture for better plant growth. Prepare fields by incorporating lime to address soil acidity and use Butachlor (10ml/litre water) as a pre-emergence weedicide. Apply balanced fertilization at 20:60:40 kg NPK/ha, split into basal and top-dressing applications. To protect crops, spray Dichlorvos (2ml/L) against Blister beetles during flowering and implement monthly preventive sprays of Dichlorvos (2ml/litre) + Bavistin (2g/L) from 30 days after sowing. Ensure light

for optimum crop performance. Harvest mature cobs promptly to prevent spoilage. For enhanced yield and soil health, adopt intercropping with soybean, kholar, or other legumes.

Management of Fall Armyworm: Sowing maize after the onset of rains reduces insect infestation. Regularly hand-pick and destroy egg masses and larvae by crushing or immersing in kerosene water. Apply sand or ash into the whorls of affected plants. Use pheromone traps @ 15/acre to trap male moths, and spray neem-based formulations (NSKE 5% or neem @ 4 ml/L) at early infestation stages. Apply a whorl spray of *Metarhizium anisopliae* @ 5 g/L at 15-25 DAS, followed by 1-2 sprays at 10-day intervals. For effective control of late instar larvae, use *Bacillus thuringiensis* var. *kurstaki* @ 2 g/L, Spinosad @ 0.3 ml/L, or prepare poison bait using rice bran, jaggery, and Thiodicarb.

MILLET

Prepare the field with 2-3 harrowings followed by leveling. Sow the crop in March to utilize early soil moisture or in August to benefit from residual moisture after the kharif season or late monsoon rains. Use a seed rate of 8-10 kg/ha, sown at a depth not exceeding 3 cm, and maintain a line spacing of 25 × 8 cm for optimal plant population and intercultural operations. Farmers are suggested to select local varieties such as Shongtak, Langjai, Buyan, Tansho, Senyak, Anong, Yempang, Shonye, and Phuknyak, or recommended improved varieties like SiA 3085 and SiA 3156 for higher productivity and adaptability. Apply 10-15 tonnes of well-decomposed FYM per hectare. Keep the crop free from weeds through one or two manual weedings during early growth stages. Treat seeds with Bavistin or Captan @ 2-3 g/kg to control seed-borne diseases. Rogue out off-type and diseased plants at flowering and maturity stages to maintain genetic purity and reduce disease spread. For neck and finger blast, spray *Pseudomonas fluorescens* twice-once at flowering and again 10 days later. Foxtail millet generally does not face major insect issues; however, field sanitation and timely removal of diseased plants will help maintain crop health.

OILSEED CROPS

Soybean

Select high-yielding soybean varieties such as JS-95-60, JS-97-52, JS-2093, RVSM-1135, and MACS-1460 for better yield potential. Sowing should be completed between mid-June and the first fortnight of July in well-drained loamy or sandy loam soils, free from water stagnation. Treat seeds with *Rhizobium* and *Phosphotika* @ 20 g each per kg of seed to improve nodulation and nutrient uptake. Use line sowing with a spacing of 45 cm between rows and 5 cm between plants for optimal plant population and crop management. Maintain a weed-free crop up to 60 days after sowing (DAS). Conduct at least two intercultural operations-first at 25 DAS and the second at 45 DAS-to support healthy crop growth and suppress weed competition.



Varietal demonstration in soyabean

25 WG (thiamethoxam) or 6.25 litre MAK HMO (Horticulture Mineral Oil) in 500 litres of water per acre in the first week of September. Even spray of Soybean oil @ 10 litres or 10 litres of cotton seed oil + 1.25 kg of detergent powder in 500 litres of water is equally effective. There should not be any water stress in the orchard at the time of spray of MAK HMO, soybean oil and cotton seed oil. During extreme hot months spray should be done in the evening hours. To manage fruit flies, regularly collect and bury the infested fruits. Fix PAU fruit fly traps @16 traps/ acre in the second week of August and recharge the same if required. Spray soybean oil @ 10 litres or 10 litres of cotton seed oil + 1.25 kg of detergent powder in 500 litres of water per acre in May-June and August-September as soon as mite population is observed on the underside of leaves. To manage Foot Rot/ Gummosis, avoid flood irrigation. Do not allow irrigation water to stay in direct contact with the tree trunk. Avoid deep hoeing. Avoid injuries to the trunk bark and roots. Give two applications of Curzate M 8 as paint (2 g/100 ml of Linseed oil) to the infected trunk portion and drench (25 g/10 litres of water/ tree) the soil at the base of the tree in July-August. During last week of June to August, apply three sprays of Bordeaux mixture (2:2:250) or 50 % Copper oxychloride (3 g per litre of water) at 20 days interval to manage scab which appears as small, dark brown, rough, irregular, raised lesions mostly on underside of the leaves, do clipping of infected leaves in July Give light and frequent irrigations in May-June and avoid water stress to combat problem of fruit cracking.

Guava

Apply Farmyard manure in May. Application of 20 kg Paddy straw manure can be used as a substitute of Farm Yard Manure to full grown guava plants. To full grown guava trees, apply urea @ 1.0 kg/plant, SSP @ 2.5 kg/plant and MOP @ 1.5 kg/plant. Half of the inorganic fertilizers should be applied in May-June and the remaining half in September-October. Correct Zinc deficiency by spraying guava trees with zinc sulphate (1 kg of zinc sulphate + 1/2 kg of unslaked lime in 100 litres of water). Give 2 or 3 such sprays at fortnightly interval between June to September. Typical symptoms of Zinc deficiency are narrowing of leaves; interveinal chlorosis and twig die back. For weed management, apply paddy straw @ 4.0 t/acre as mulch after application of recommended dose of organic manure and inorganic fertilizers in May under the tree canopy. Guava gives two crops in a year. Winter season crop is superior in quality than rainy season crop, as rainy crop may get infested with fruit flies. Those farmers who intend to get only winter season crop, can spray urea 10 per cent (100 g



Fruit Fly in Guava



Anthrachnose in Guava

Soybean

Time of Sowing: Soybean sowing should be sown by 15th June where irrigation facility is available. If irrigation facility is not available, sowing should be done with onset of monsoon.

Improved Varieties: Improved varieties are- NRC 138, RVSM 2011-35, JS 20-116, JS 20-98, JS 20-94, JS 335, NRC 127, JS 93-05, Pratap Soya 1, Pratap Soya 2 (RKS 18), Pratap Raj 24 (RKS 24), JS 97-52, JS 95-60, Pratap Soya 45 (RKS 45), JS 20-34, JS 20-29 and RKS 113

Seed Rate: 80-100 kg/ha.

Spacing: Row-to-Row spacing 30-45 cm and plant to plant 10 cm.

Seed Treatment: Treat the seeds with Carbendazim or Thiram @ 2g/kg of seed 24hrs before sowing or with talc formulation of *Trichoderma viride* @ 4 g/kg seed (or) *Pseudomonas fluorescens* @ 10 g/kg seed.

Fertilizer Application: Apply 20 kg N, 40 kg P₂O₅ phosphorus and 40 kg K₂O per ha before sowing. If the seeds not inoculated with *rhizobium* culture than use 40 kg nitrogen per ha. To get higher yield of soybean foliar spray of water-soluble N:P: K (19:19:19 or 17:44:00) @ 0.5% (5 g per litre of water) at 45 and 60 days after sowing is recommended. Soil incorporation of 30 kg sulphur/ha through gypsum before sowing and 1.0 kg boron per ha (borax @ 5 kg/ha) in boron deficient soil should also be done.

Water Management: Flowering and pod formation stage in soybean are the most critical for water stress, 1-2 irrigations should be applied as per the need.

Weed Management: Pre emergence application of Pendimethalin 30 EC at 1.0 kg a.i. per ha (commercial dose 3.3 liter per ha) and one hand weeding at 30-35 days after sowing. Post emergence application of Sodium Acifluorfen 16.5 % + Clodinafop Propargyl 8 % EC (ready mix) at 1000 ml per ha at 20-25 DAS effectively control the narrow as well as broad leaf weeds.

Plant Protection: Stem Fly: to manage apply Quinolphos 25 EC @ 1.5 litre per ha or Thiamethoxam 12.6% + Lamdacyhalothrin 9.5% ZC @ 125 ml per ha. Griddle Beetle: can be managed by application of Dimethoate 30 EC @ 1.0 litre per ha or Thiacloprid 21.7 SC @ 750 ml per ha. Defoliators (Green semilooper, Tobacco caterpillar & gram pod borer): can be managed by application of Spinetoram 12 SC @ 450 ml or Quinolphos 25 EC @ 1.5 litre or Emamactin benzoate 5 SG @ 180 gm or Indoxacarb 15.8 EC @ 300 ml per ha. Seedling Rot: The disease results in poor seedling emergence because of seed rotting. The seed treatment with Thiram at 3.0 g/kg of seeds is beneficial. Pod Blight: The pods of affected plant turns yellowish green and later on dry up. The grain becomes shrivelled and mould. Use of resistant varieties, sowing of crop in rotation and spray of 0.5 per cent Zineb can control the disease.

Yellow Mosaic Transmitted by white fly (*Bemisia tabaci*). The disease is dangerous and widely spreads. Spray Metasystox 0.2 per cent at 10-15 days interval controls the vector effectively.

Sesame (Til)

Time of Sowing: Sowing with onset of monsoon and to be completed before mid-July. Crop is highly sensitive to water logging conditions; therefore, proper drainage facilities are must for higher productivity.



Farmers are advised to follow the sowing rule to take-up sowing of rainfed crops like Cotton, Soybean, Maize, Jowar, Redgram, Greengram etc., only after receipt of cumulative rainfall of 50-60 mm in light soils and 60-75 mm in heavy soils or the soil should be wet up to top 15-20 cm depth after enter of South-West monsoon rains into the state.

CEREAL CROPS

Paddy

Under assured irrigation farmers are advised to raise rice nurseries of long duration (140-145 days) varieties from May 20th to June 20th. Under assured irrigation farmers are advised to raise rice nurseries of medium duration (130-135 days) or short duration (120-125 days) varieties from June 20th onwards. Month of June is not optimum for raising paddy nurseries using Telangana Sona (RNR15048) variety. Take the Greengram as catch crop to preceding rice in areas where the release of water into the canals and tanks is delayed. Green manuring with Sesbania/ Crotalaria / Pillipesara / grain legumes, residues of blackgram / greengram and in situ incorporation before flowering. Varieties: cultivation of long duration (BPT-5204, WGL-44, RNR 11718 etc.), medium duration (WGL-14, JGL-11470, JGL-384, RNR-28361, JGL-33124 etc), short duration (RNR-15048, KNM-1638, KPS-6251, WGL-362, KNM-118, JGL-24423, RNR-29325, JGL-28639 etc.) varieties.

Direct seeding of rice with Seed-cum ferti drill wherever suitable. In Telangana state, dry converted wet and wet direct seeded rice are popular, and the sowing can be started from June ending. Treat seeds with carbendazim @ 3 g/ kg (dry seed treatment) or 1 g/L (wet seed treatment). Farmers are advised to apply the Carbofuran 3CG granules @ 800 g per acre of nursery one week before pulling of the nursery. Root dipping: prepare the slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000 g/ha) of phosphobacteria or 10 packets of (2000 g/ha) of Azophos, inoculant in 40 lit. of water and dip. the root portion of the seedlings for 15-

30 minutes in bacterial suspension and transplant. Make 20 cm alleyways at every 2 meters to facilitate free aeration, ease of farm operations and better pest and disease management. Broadcast butachlor @ 1-1.5 l or pretilachlor @ 500-600 ml or 4 kg granules of bensulfuron methyl (0.6%) + pretilachlor (6.0%) with 20 kg sand per acre. Practice alternate wetting and drying in well irrigated areas to save water, better aeration and minimize incidence of BPH besides getting carbon credits with less emission of greenhouse gases. Recommended dose of fertilizers: 40-48 Kg Nitrogen, 20 Kg Phosphorus 16 Kg Potash per acre during Vanakalam. Spray 0.2% ZnSO₄ 2 to 3 times at 5 days interval to manage Zn deficiency, particularly in alkaline soils.

Maize

Deep summer plough to destroy pests and conserve moisture. Optimum sowing window is from June 15th to July 15th with medium and long duration varieties (90-100 days and 100-120 days) without any yield reduction. Sow medium duration varieties/hybrids DHM-206, DHM-182, DHM-117, Bisco-855, Bio-9637, Bisco-740, Kaveri Ekka-2288, Kaveri -2020, Kaveri-3696 and short duration hybrids DHM-115, Pioneer-3342, KH-5991, DKC-7074 during 15th June to 15th July. Sowing can be extended up to July end if monsoon is delayed. Sow on receipt of at-least 50-60 mm rainfall for better germination and complete sowing in a week for reducing the incidence of FAW. Spray NSKE 5% or neem oil 1500 ppm @5ml/l after sowing and at weekly intervals. Dibble seeds with tractor drawn cultivator or seed cum ferti drill on ridges and beds. Intercrop red gram varieties PRG-176, WRG-97 in 2:1 ratio in light soils and in 4:1 ratio in heavy soils to reduce incidence of FAW. Follow IPM for FAW management: summer ploughing, growing single cross hybrids, clean cultivation, intercropping with pulses, balanced fertilization, release of egg parasitoids (*Trichogramma* sp) and spraying Chlorantraniliprole @ 0.4 ml or spinetoram @ 0.5 ml or Emamectin benzoate @ 0.4 g per litre in plant whorls. As post emergence, spraying of Tembotrione @ 115 ml + Atrazine 400 g or Toperamezone @ 40 ml + Atrazine 400 g or mesotrione + atrazine @1400 ml/acre in 200 litre of water will effectively control weeds. After 30-35 days, crop may be inter-cultivated followed by earthing up is needed for better nutrient uptake and to avoid lodging.

COMMERCIAL CROPS

Cotton

Sow the crop early by June 1st/2nd week after receipt of sufficient good rains at least (60mm). Don't sow cotton crop after July 20th as yields will be reduced drastically under rainfed conditions. Sowing of Cotton on ridge and furrow in medium soil and broad Bed furrow in black soil under rainfed conditions facilitates the crop to moisture conservation and excess water drain out. In a village or area all the cotton cultivating farmers should take up the sowings on community basis and complete within a period of 7 - 10 days. Depending on soil type and rainfall pattern cotton should be sown at a spacing of 120 x 60 cm/ 120 x 45 cm/ 90 x 60 cm/ 90 x 45 cm/ 90 x 30 cm. Deep summer plough to destroy the eggs of insect pests and to remove weed stock. Intercrop with greengram/ blackgram/ soybean/ cluster bean in 1:2 or 1:3 ratio, and red gram 4:1 / 6:1 / 8:1 ratio which also helps to manage sucking pests.

Grow 2-3 rows of maize or sorghum as border crop as barrier to prevent pests. Treat the seed with imidacloprid 70 WS@ 5 g per kg or thiamethoxam 70% WS @ 4 g per kg followed by *Trichoderma viridi* or *Pseudomonas fluorescens* @ 10 g per kg.

Pre-emergence herbicides like pendimethalin (Stamp extra) @ 700 ml/acre should be applied within 24 -48 hours after sowing with the help of hand sprayer for better control of weeds in cotton. Application of post emergence herbicides like quizalofop ethyl @ 400 ml + pyrithiobac sodium @ 250 ml in 200 L of water at 30 DAS in cotton to control grasses and broad-leaved weeds. Inspect the crop at squaring and flowering stage for presence of PBW larvae within flowers. Installation of Pheromone traps @ 8/acre and monitoring ETL at this stage is 10 % damaged flowers. Spray $MgSO_4$ @ 10 g/l twice at 45 and 75 DAS; Boron @ 1.5 g/l twice at 60 and 90 DAS; 0.5% $FeSO_4$ with citric acid twice at weekly intervals and $ZnSO_4$ @ 2 g/l twice at 4-5 days interval at 45 DAS to manage micronutrient deficiencies. During excess rainfall, drain excess water and apply 25 kg urea and 20 kg potash per acre. Spray Planofix (NAA 10 PPM) @ 1 ml per 4.5 liters of water twice at 10-15 days interval to manage flower dropping. Spray 1% KNO_3 or 19:19:19, 20:20:20, 21:21:21. Follow rotation of insecticide sprays against insect pests for high efficiency. Stem application of fipronil 1: 4 at 30 DAS and Flonicamid / Imidacloprid with water (1:20) at 45, 60 DAS to be done which protects the crop from early season sucking pests like jassids and aphids.

PULSE CROPS

Pigeon pea

Sow high-yielding wilt-resistant varieties (155-165 days) like WRGE-93, WRGE-97, WRGE-121, and medium-duration varieties like WRG-255 and TDRG-4, TDRG-59 & TDRG-272 can be recommended. Sowing should be initiated only after receipt of 60 mm rainfall. The optimum sowing time should be 15th June to 15th July at a spacing of 150-180 x 20 cm in black soils and 90-120 x 20 cm in red soils. Seed rate should be 2.5 to 3.0 kg per acre only (plant-to-plant spacing should be mandatory). Treat seeds with thiram or captan @ 3g/kg followed by rhizobium culture @200-400 g/ 3 kgs of seed. The raised bed method or ridge and furrow method should be followed to protect the crop from *Phytophthora* blight disease, which occurs due to excess and stagnation of rainwater. Intercropping in Red gram + Cotton (1:4/ 1:6), Soybean (1:4/ 1:6), Green gram/ Black gram (1:7) should be followed. Treat the Red gram seed with *Trichoderma viride* @ 10 g/ kg seed to protect the crop from soil-borne diseases.

Spray pendimethalin @ 1 - 1.5 L in 200 liters of water within 24 - 48 hours after sowing will effectively control the weeds in Red gram Post-emergence application of quizalofop ethyl @ 400 ml/ acre to control grasses and imazethapyr 10 % @ 300 ml/ acre to control grasses and broad-leaved weeds. Both herbicides in the tank mix are also recommended in case both weeds are observed. If there is submergence due to excess rain, drain excess water at the earliest and spray 1% KNO_3 or 19:19:19, 20:20:20 or 21:21:21. Application of Multi K or 13-0-25 @ 10g/l at the time of flowering should be done. Practice nipping at 45 and 60 DAS, wherein terminal branches are removed to facilitate lateral branches, thus improving yields. To control pod fly spraying of Thiomethaxam 25 wg @ 0.2ml/l, Flubendamide 0.2ml/L with Jaggery 5g/L to be sprayed at the time of early pod formation stage.

Greengram and Blackgram

Sowing time is generally from June 15th fortnight- July 15th during Kharif, Sept 15th to 30th October during Rabi, January last week to March 15th during Summer. Greengram varieties MGG-295, MGG-385, MGG-351, TM-96-2, WGG-37 and WGG-42, Virat, IPM-2-14, Sikha. Blackgram YMV-resistant varieties MBG-1070, PU-31, GBG-1 and TBG-104 are the preferred varieties. Treat seeds with imidacloprid @ 5g or thiamethoxam @ 5g/kg. Intercropping can be done with cotton (2:1 or 1:1 ratio) or redgram (6:1)/(8:1). Greengram can be taken as a preceding crop to Rice and Chilli as green manure for incorporation in the soil.

OILSEED CROPS

Groundnut

Seed rate recommended is 237 kg/ha and practice seed treatment with imidacloprid 600 FS @ 2 ml/kg followed by tebuconazole 2 DS @ 1g or Trichoderma viride @ 10g/kg to control early and late leaf spot diseases, sucking pests, collar rot, PSND in the early stages. Application of Gypsum @ 500 kg/ha at pegging stage near podding zone for pod development. Weed management with Diclosulam 84% WDG @31 gm/ha as pre-emergence herbicide followed by post-emergence application of Imazethapyr @ 750 ml/ha at 20 to 25 days after sowing.

Soybean

Sow varieties ASB-22, JS-335, AISb-1, JS-93-05 during 15th June 1st week of July using seed drill/ gorru/ desi plough/ by tractor drawn seed cum fertilizer drill / broad bed furrow planter. Sowing of Soybean by Broad bed Furrow (BBF) planter or Raised bed method facilitates the crop to withstand heavy rains and moisture stress conditions and to obtain higher yields. Treat the seeds with carbendazim @ 1g or thiram or captan @ 3 g/kg followed by 1.5 ml of imidacloprid (48 %) and Rhizobium culture @ 200 g/10 kg seed. Apply 50 kg DAP + 26 kg MOP at the time of sowing and top dressing of urea @ 25 -30 kg/acre at 30 DAS. Pre-emergence herbicides like pendimethalin 30 % @ 1 - 1.5 L/ acre should be applied within 24-48 hours after sowing with the help of hand sprayer for better control of weeds. Post emergence application of quizalofop ethyl @ 400 ml/ acre for control of grasses and imazethapyr + imazomax 40 g /acre or imazethapyr 10 % @ 250 ml/acre to control the grasses and broad-leaved weeds. To control stem fly and stem gridler as a prophylactic application of carbofuran 3G granules @ 8 kg/acre is recommended at 10 DAS. To control of pod blight spraying of tebuconazole + sulphur @ 2.5 g/L is recommended. Dry the seed and maintain the below 12 % moisture during seed storage to avoid incidence of pest and disease.

Castor

Sow high yielding castor hybrids viz., ICH-5, ICH-66, PCH-111, DCH-519 & ICH-6, PCS-262 from 15th June to 15th July in rainy season under rainfed conditions. Intercrop with redgram in 1:1 or greengram in 1:3 ratio, open furrows between rows to conserve moisture in light soils.

LIVESTOCK

Animal Husbandry & Fisheries

Agriculture and allied sectors play a vital role in the lives and livelihoods of the majority of Telangana's population. The state is rich in livestock resources and

liquids & probiotics in drinking water during summer. Utilize broken rice or bajra (up to 30-40%), cotton seed meal (10-12%) and rapeseed meal (5 -8%) in poultry when there is shortage of maize during natural calamities. Keeping in view of disease outbreak follow biosecurity measures and mass vaccinate backyard poultry birds against Newcastle and viral diseases to prevent outbreak during stress periods. For hatching purpose of fertile eggs keep it in fridge at temperature of 4°C. To improve the growth rate 1 per cent protein level (soybean meal) may be added in the feed. Suitable coccidiostat should be added in the feed continuously by consulting the local veterinarian to prevent coccidiosis. 3-5 g of oyster shell/limestone/grit per day/bird shall be given to laying hens to avoid leathery eggs.

Value Addition: Convert milk into ghee, paneer, curd, khoa for better returns. Explore meat processing and packaging for hygienic sale. Utilize manure for biogas and compost.

Fisheries

Failed Monsoon: Use bore well or alternative water sources to maintain minimum pond water levels. Reduce stocking density to lower water demand. Adopt partial harvest strategies to minimize losses. Use water-saving practices like reduced water exchange. Avoid new stocking if water availability is uncertain.

Flood: Strengthen pond bunds and spillways to prevent overflow and escape of fish. Install protective netting around pond perimeters. Avoid feeding during peak flood to prevent water quality issues. Harvest fish early if flood risk is imminent. Apply disinfectants like Benzalkonium Chloride 80% (BKC) @ 2.5 L/ha and Rock Salt @ 75-125 kg/ha/5ft depth/ha after floodwater recedes to control disease outbreaks.

Drought: Reduce biomass load in ponds by early or partial harvesting. Avoid use of organic manure if water level is too low to prevent oxygen depletion. Apply aeration measures (e.g., paddle wheel aerators) to maintain DO levels. Maintain good water quality by regular monitoring and minimum exchange.

Delayed Monsoon: Postpone seed stocking until adequate water level is available. Increase application of manures/fertilizers once rains begin to promote plankton growth. Avoid breeding activities during uncertainty to protect brood stock. Maintain brood stock in protected tanks.

Advanced Monsoon: Prepare ponds early for stocking, complete liming 200 kg/ha and fertilization in advance. Ensure brood stock is not disturbed during monsoon (15th June to 31st August). Avoid overfeeding as heavy rains can cause feed waste and water quality degradation. Watch for sudden temperature or pH fluctuations due to early rains.

High Temperature: Maintain higher water levels (5-6 feet) to buffer temperature fluctuations. Avoid feeding during peak heat hours, feed early morning and late evening. Use shade nets or plant trees around ponds to reduce direct sunlight. Monitor dissolved oxygen (DO) and apply aeration during hot afternoons.

Low Temperature: Reduce feeding rate and switch to highly digestible feed. Avoid handling fish during early mornings and nights. Provide windbreaks and increase pond depth to retain heat. Avoid use of antibiotics and strong chemicals during cold periods unless essential.

SPICE CROPS

Ginger & Turmeric

Farmers are suggested for planting ginger (Varadha, Nadia) and turmeric (Megha Turmeric 1 or Lakadong) rhizomes in the first week of May for optimal growth. Before planting, dip the rhizomes in *Trichoderma Viride* (4g/L of water for 5 minutes) or *Trichoderma Harzianum* (10ml/L) to prevent soil-borne diseases. Additionally, spraying Neem oil (10000ppm @ 2ml/L) or *Trichoderma* solutions can help prevent soft rot. Apply farmyard manure (FYM) at 25t/ha, and fertilizers with a ratio of N:P:K 75:50:50 kg/ha for better growth and yield. Plant rhizomes in ridges and furrows with a spacing of 40x20 cm. Ensure proper weeding and earthing up. Intercrop with crops like soybean, French bean, finger millet, or maize to improve farm income. Mulch the beds with dry paddy straw or green leaves to conserve soil moisture, control weeds, and prevent soil erosion.

LIVESTOCK

Regularly de-worm livestock as per their body weight and condition, especially during the hot and humid kharif season when worm infestations are more prevalent. Keep the livestock shed clean by using potassium permanganate for regular disinfection, and apply phenol compounds outside the shed to reduce the multiplication of disease-carrying vectors like mosquitoes and flies. Begin planting perennial fodder varieties like Nappier, Congo Signal, and Moringa. Additionally, prepare for the lean season by drying maize stalks for hay or making silage. Vaccinate cattle against FMD, HS, and BQ, goats over three months old against PPR, and pigs against Swine Fever. Apply external ectoparasiticides to prevent ticks, mites, and fleas. Keep the poultry shed dry, apply lime regularly to reduce the incidence of coccidiosis, and ensure proper ventilation to prevent disease outbreaks. Restrict poultry from exposure during rainy or stormy weather to protect from seasonal fever and other environmental stress factors. Use herbal or chemical fly repellents on a daily basis to manage external parasites and prevent fly infestations in the poultry farm.

FISHERIES

It is recommended to focus on carp breeding, hatchery management, and carp fry and fingerling production from April to July, as this is the peak time for these activities. For optimal carp fingerling production, dewater seasonal ponds to remove unwanted fish. Plough the pond bottom, expose it to sunlight for one week, and apply lime at a rate of 400 kg per ha to prepare the pond. After a week, apply cow dung at 10,000 kg per ha. Two weeks later, remove aquatic insects by using soap and kerosene emulsion. The exact doses should be confirmed by consulting a fisheries expert. Once the pond is prepared and insect removal is complete, stock healthy 4-day-old carp spawn at a density of 30 lakh/ha. This should be done 1-2 days after insect removal. Ensure regular feeding of carp fry with a mixture of finely powdered MOC and RB powder in a 1:1 ratio to promote healthy growth. It is advisable to excavate or renovate water harvesting ponds to conserve rainwater and runoff, maximizing water storage during the monsoon season. Maintain pond embankment stability by ensuring side slopes of 2:1 (Horizontal: Vertical) for sandy soil, 1.5:1 for loamy soil, and 1:1 for clayey soil. Additionally, create an empty berm of a width equal to the depth of the pond to prevent siltation during the rainy season. To maintain water quality, check the pH level of the pond after heavy rainfall. If the pH falls below 6.5, apply lime at 50 kg/ha to improve the water's alkalinity.

cultivated in the field for the first time or after a long gap, to enhance nitrogen fixation. For weed control, one or two hand weeding should be done up to 40 days of sowing depending upon the weed intensity. Weeds can be controlled by the use of Pendimethalin 0.75-1.00 kg a.i./ha in 400-600 L of water as pre-emergence application.

OILSEED CROPS

Groundnut

Cultivate improved groundnut varieties such as Chitra, Kaushal, Prakash, Ambar, TG-37A, Utkarsh, and Divya, based on the length of the growing season and end-use requirements. Groundnut matures in about 120-130 days and should ideally be sown in the first fortnight of July to ensure harvest by the end of October.

Maintain row spacing of 45-50 cm and plant-to-plant spacing of 15-20 cm. Use 70-75 kg seed kernels/ha for timely sowing, and 90-95 kg/ha for late sowing conditions. Treat seeds before sowing with Thiram @ 2 g and Carbendazim @ 1 g per kg of seed, or Trichoderma @ 4 g + Carboxin @ 1 g per kg. Inoculate seeds with groundnut-specific Rhizobium culture 5-6 hours before sowing. Dissolve 50 g jaggery in 0.5 liters of water, mix with 250 g Rhizobium culture, and coat over 10 kg of seed. Dry treated seeds in the shade for 2-3 hours and sowing of seeds should be done in early morning or evening times to protect Rhizobium from heat-induced mortality.

Apply fertilizers based on soil testing; in the absence of test results, apply 20 kg nitrogen, 30 kg phosphorus, 45 kg potash, 250 kg gypsum, and 4 kg borax per hectare. Perform two hoeing-cum-weedings, the first at three weeks after sowing and the



Groundnut (TG 37A) at Mainpuri



Sesame (GJT 5) at Amethi

second three weeks later, as this helps minimize early weed competition, promotes better aeration, and supports healthy crop growth. For additional control, especially under high weed pressure, apply pre-emergence herbicide imazethapyr @ 100 g/ha in 500-600 liters of water within 3 days of sowing.

Soybean

Cultivate improved soybean varieties such as PK-472, JS-715, PS-564, PK-262, JS-2, JS-935, JS-7244, JS 20-29, Pusa-20, Pusa-16, and PK-416 based on growing season and production goals. The crop matures in about 120-125 days; sow between mid-June and the first fortnight of July. Maintain a row spacing of 45 cm and plant-to-plant spacing of 3-5 cm; sow seeds 3-4 cm deep. Use 70-75 kg seed/ha under normal conditions and 75-80 kg/ha under late sowing. Ensure a minimum germination of 75-80%.

plant hopper and gandhi bug management. Spray of hexaconazole 5%EC 1000ml or azoxystrobin 18.2%+ difenconazole 11.4% SC 500ml or mancozeb 75WP 1.5-2.0 kg or isoprothiolane 40% EC 750ml/ha for of blast and sheath blight disease management in rice.

Maize

Recommended sankul varieties for Tarai, Bhabhar & Plain area are Pant Sankul Makka-3, Sweta, Bajora Makka-1 & Vivek Sankul-11. Hybrids maize are H M 4, 10, 11, 13, HQPM 1, 4, 5, Pant Sankar Makka-1, 4, 5, 6, Sartaj, P-3522 and Vivek Makka-51. Best sowing time is mid-June in Uttarakhand plains. Popcorn varieties: Pant Popcorn-1 and VL Amber Popcorn, fodder maize variety is African Tall. Sowing time is mid-June to mid-July. Sankul varieties suitable for hills: Vivek Sankul Makka-11, 37, 31, 35 and Pant Sankul Makka-4, hybrids maize are Vivek Sanker-9, 39, 45, 53, 55, 57 and DMH-109. Time of sowing in lower hill (1st week June to mid-June), mid hill (May end to mid-June) and higher hill during April end to mid-May is suggested. Optimum seed rate of 18-20 Kg for sankul, 20-22 Kg for hybrids, 12-14 Kg for popcorn, 40-45 for baby corn and 8 kg/ha for sweet corn is recommended. Spacing for maize may be maintained at 60*20-25 cm in line sowing at a depth of 5 cm. Inter-cropping of cowpea/soybean/ginger in maize would be effective to achieve the higher productivity, profitability and also control the soil erosion in hills.

Fertilizer Application: The maize crop in Tarai, Bhabhar & Plain area should be fertilized with NPK @ 120:60:40 kg/ha. Full P and K, $\frac{1}{4}$ N apply at sowing and rest of N applied in 3 equal splits at 3-4 leaf stage, knee stage and flowering time. In case of zinc deficiency symptoms in previous crop, apply zinc sulphate (21%) 25 kg/ha as basal application.

Weed Management: For proper weed control minimum two hand weeding at 20-25 and 40-45 days after sowing is recommended. Application of atrazine 2.5 kg/ha as pre-emergence for broad leaf weeds fb tembotrione 42% SC 120 g/ha post-emergence at 3-4 leaf stage (15-20 DAS) for grassy and broad leaf weeds or alachlor 2.0 kg/ ha as pre-emergence for both grassy and BLW is recommended. To control the sedges, application of halosulfuron methyl 70WG 67.5g a.i./ha as post-emergence at 3-4 leaf stage (15-20 DAS) is recommended. For control of broad leaf weeds in maize crop application of 2, 4 -D dimethyl amine salt 58% SL 0.5 kg/ha or 2, 4-D ethyl ester 38% EC 0.9 kg/ha as post-emergence at 15-20 DAS.

Plant Protection: Banded Leaf and Sheath Blight disease in maize can be effectively managed through an integrated approach. Soil application of Trichoderma (Pant Bioagent-1) at the rate of 1 kg multiplied in 100 kg of vermicompost (VC) or farmyard manure (FYM), applied 10-15 days prior to sowing, has shown good results. Additionally, seed treatment with Trichoderma (Pant Bioagent-1) at 20 g



FAW infestation in Maize

Potato

Recommended Varieties of potato for hills are Kufri Himalini and Kufri Jyoti with seed rate of 1.5 to 2.5 t/ha. Recommended crop geometry for potato cultivation is 50 cm X 20 cm. During last ploughing around 10-15 tonne of FYM plus basal dose of 90 Kg Urea, 300 Kg SSP and 65 kg MOP is recommended to be applied followed by top dressing of Urea at the rate of 70 kg/ha and 50 kg/ha 30 and 50 DAS in potato cultivation. Earth up around the base of plants to encourage better tuber formation, particularly when plants attain 15-20 dm height. Hand weeding, followed by earthing up and mulching with local material like Baanj leaves should be performed for effective weed management. In plains of Uttarakhand, where chemicals are used, Pre-emergence application of Metribuzin 70% WP at the rate of 250-300 g/acre or post emergence application of Propaquizafop 10% EC at the rate of 2 ml/L water or 400 ml/acre are found some excellent choices to control weed flush.

Plant Protection Practices: Larvae of Potato tuber moth, cause damage in the form of mines in the leaves and/or weakening of the stem, which can break. To control/manage this pest, avoid planting potato tubers in shallow soil. Plant the tubers at a depth of 10 to 15 cm. for biological control, install pheromone traps 20/ha and spray NSKE 5% or Quinalphos 20 EC 2 ml/L (ETL 5% leaf damage) to control foliar damage. Potato Cut worm, is one of the most damaging pests in rainfed areas of Uttarakhand. To manage this pest, set up a light trap to attract adult moths during summer. Install a sprinkler irrigation system to potato farm and irrigate the field during the day to expose the larvae to bird predation. As chemical control, a day after planting, drench the collar region of the plants with chlorpyrifos 50% + cypermethrin 5% at the rate of 1ml/L of water. To manage White Grub, go for summer ploughing to expose pupae and adults, install light traps and should be active between 7 PM. to 9 PM Major diseases of potato in Uttarakhand are Late blight and Early blight. To manage these two diseases, remove any ground creepers that are a source of infection. Spray Mancozeb 2 g/L water or Chlorothalonil 2.5 g/L water at 45, 60, and 75 days after planting is recommended for late blight and Difenoconazole 25% EC or Hexaconazole 5% EC at the rate of 2ml per litre of water for early blight.

BEEKEEPING PRACTICES DURING RAINY SEASON

Colonies should not be inspected on cold, windy, and cloudy days, and during heavy rain, hives should be protected with roofing sheets. Always use protective dress and veils while inspecting colonies, handle them gently without jerks, and isolate diseased colonies from healthy ones. Ensure the provision of fresh water nearby in shallow vessels, and provide 50% sugar syrup during dearth periods when nectar and honey stores are inadequate. Avoid feeding colonies with honey or syrup mixed with honey. When pollen stores are low, provide pollen substitutes made of fat-free soybean flour (3 part), brewer's yeast (1 part), skimmed milk powder (1 part), sugar (22 parts), and



Wax moth infestation in Bee Hives

When pollen stores are low, provide pollen substitutes made of fat-free soybean flour (3 part), brewer's yeast (1 part), skimmed milk powder (1 part), sugar (22 parts), and

Field Crops

India has one of the largest agricultural sectors worldwide, which is the primary source of livelihood for ~55% of India's population. India has the world's largest cultivated land under wheat, rice, and cotton. India is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. The agriculture sector in India holds the record for second-largest agricultural land in the world, generating employment for about half of the country's population. Thus, the farming community comprise an integral part of this sector to provide food security with a means of sustenance.

During the last 9 years (2014-2023), a total of 2279 varieties of field crops have been released in India under the aegis of ICAR, which include 1092 of cereals, 343 of oilseeds, 343 of pulses, 137 of forages crops, 279 of fibrecrops, 68 of sugarcane and 17 of potential (underutilized) crops. These varieties include 1888 climate resilient varieties (83%) tolerant to one or more biotic and/or abiotic stresses. Out of these 1888 varieties tolerant to one or more biotic and/or abiotic stresses, 407 varieties have been developed specially for extreme climate including flood/ water submergence/ water logging tolerance (73), drought/moisture stress/ water stress tolerance (220), salinity/ alkalinity/ sodic soils tolerance (52), heat stress/ high temperature tolerance (49), cold/ frost/ winter chilling tolerance (13) using the precision phenotyping tools. In horticultural crops also, 22 climate resilient varieties have been developed. This includes varieties of potato (5), tomato (2), spinach (1), radish (1), cassava (4), sweet potato (2), greater yam (1), white yam (1), coconut (3) and taro (2).

Further, to tackle the widespread problem of malnutrition in globally, concerted efforts have been made to develop varieties rich in various essentially required micro-nutrients (Protein, Fe, Zn, Vit. A, Vit. C, oleic acid, anthocynin, lysine, tryptophan, removal of anti-nutritional factors like erucic acid, glucosinolates, KTI) for proper growth and development of the body and as a result 124 field crops and horticultural crops varieties have been released and notified during past nine years [Rice (10), Wheat (41), Maize (20), Pearl millet (9), finger millet (4), little millet (1), Lentil (2), Mustard (8), Linseed (1), Soybean (7), groundnut (3) and 18 Horticultural crops]. Concerted efforts for seed production have been made to take the biofortified varieties in general cultivation, which have resulted in coverage of more than 10 million hectares during 2022-23. Biofortified varieties have more than 50% share in total wheat breeder seed indents.

1. Implementation of GAP

The ministry of Agriculture and Farmer's Welfare and Processed Food Products Export Development Authority (APEDA), Govt. of India implements the GAP. APEDA adopts the approach of developing product specific clusters in different agro- climatic zones of the country to help in dealing with various supply side issues through Good Agriculture Practices addressing issues such as, soil nutrients management, higher productivity, adoption of market-oriented varieties etc.